James Sypherd

sypherdj1@gmail.com | 602-321-4399 | jamessypherd.com | github.com/BetterCodeGoblin | linkedin.com/in/james-sypherd-39b30a26a

PROFILE

Biomedical engineer, researcher, and game systems designer whose work spans clinical and orthopaedic research, motion capture and biomechanics, research instrumentation, real-time interactive systems, and applied AI. Experienced building gameplay in Unreal Engine 5 and Unity, designing behavioral and measurement hardware, and developing LLM-driven tools with prompt design, guardrails, and evaluation. Comfortable moving from study design and data analysis to shipping working software.

EDUCATION

Arizona State University - M.S., Biomedical Engineering, May 2025 (GPA 3.50)
Arizona State University - B.S.E., Biomedical Engineering, May 2024 (GPA 3.60)

RESEARCH EXPERIENCE

Phoenix Children's Hospital
Clinical Research Coordinator Fall 2025

- Coordinate clinical research studies, including participant enrollment, study documentation, and data collection and management.

Phoenix Children's Hospital
Orthopaedic Researcher 2023 - 2025

- Conducted orthopaedic and movement research supporting clinical motion analysis, data processing, and biomechanical assessment.
- Contributed to motion capture and gait analysis workflows in the Motion Analysis Laboratory.

Geometric Media Laboratory, Arizona State University
Graduate Researcher, Motion Capture Fall 2024 - Spring 2025

- Built MATLAB tools to import motion capture data and run inverse kinematics analysis on OptiTrack recordings.
- Designed capture and analysis workflows for studying timing, movement quality, and biomechanical variation.

Barrow Neurological Institute
Research Intern Summer 2021

- Designed and built a capacitance-based lickometer instrument for behavioral neuroscience experiments.
- Delivered a working device for licking-response measurement, programmed on Arduino (open-sourced as Cap-Lickometer).

TEACHING EXPERIENCE

Arizona State University
Professor of Practice, GAME School, and Graduate Teaching Assistant, AME 294 Spring 2025 - Present

- Teach a 3-credit course on motion capture, movement analysis, and real-time 3D visualization.
- Help students turn physical motion into readable interactive systems using OptiTrack workflows.
- Write instructional materials and technical documentation for creative technology pipelines.

ENGINEERING & GAME DEVELOPMENT EXPERIENCE

Endless Games and Learning Lab, Arizona State University
Software Engineer / Game Developer December 2025 - Present

- Develop gameplay systems and prototype features for educational and entertainment projects across Unity, Unreal Engine 5, UEFN, and Roblox.
- Implement player-facing mechanics, interaction systems, and rapid prototypes for combat, progression, and experiential design goals.
- Apply AI-assisted prototyping and tooling to move from design intent to working mechanics faster.
- Support motion capture and .c3d data workflows for gameplay, animation, and interactive media.
- Contribute Unreal Engine 5 work on multiple in-house projects, including gameplay prototyping and mechanic implementation.

Hunting Grounds Board Game Development Team
Lead Game Designer Spring 2023 - Summer 2023

- Led design of a modular tabletop game built around procedural learning through play.

- Built a prototype with 200+ cards, class-specific interactions, progression structures, and role-based mechanics.
- Designed itemization, rewards, and player guidance to strengthen replayability and class differentiation.

AI & LLM SYSTEMS

Applied AI Work (Independent / OpenClaw agent platform)
Builder / Systems Designer 2022 - Present
- Design system prompts, role-scoped behavior definitions, and safety guardrails that keep an autonomous agent in character and within policy.
- Built a multi-provider LLM routing layer across OpenAI, Anthropic, Google Gemini, and Groq, choosing models per task by capability, latency, and cost.
- Implemented retrieval over a structured memory and knowledge store so responses stay grounded in prior context, a working RAG pattern.
- Iterate on prompt versions and check for regressions when prompts or models change, an informal evaluation and quality practice.
- Wrote an institutional audit of AI tools for Arizona State University covering capability, data handling, and approved-use paths.
- Automate data workflows, scheduling, and tool integrations in Python.

SELECTED TECHNICAL PROJECTS

Hunting Grounds (Unreal Engine 5.8, C++)
Author 2026
- Building a UE5.8 C++ adaptation of the Hunting Grounds tabletop game, scaffolded on the Gameplay Ability System, Gameplay Tags, and Enhanced Input.

Rebuild Orchestrator (Unreal Engine 5.8 Plugin)
Author 2026
- Built an editor-only C++ plugin that runs a safe external clean, build, and relaunch loop through Unreal Build Tool, verified on the StrengthERG project.

Multi-Engine Game Prototypes
Solo Developer 2024 - Present
- Built game prototypes in Unreal Engine 5 and assorted Roblox (Luau) titles.

GAME DESIGN EXPERIENCE

The Blacksmith
Gameplay & Systems Developer 2025 - Present
- Build crafting and forging gameplay systems in Unreal Engine 5, including Blueprint-driven crafting and item-creation flows.
- Create and author enemy AI behavior trees, animation montages, blackboards, and AI controllers.
- Collaborate through a Perforce-managed multi-developer pipeline, coordinating streams, workspaces, and asset submissions.

Concept 2 Exercise Game Suite (StrengthERG)
Gameplay Engineer 2024 - Present
- Build exercise games in Unreal Engine 5.8 that integrate Concept 2 PM5 rowing and erg hardware in real time over TCP and CSAFE; led the suite's migration from an original Unity build.
- Design movement-driven mechanics that map stroke data and heart rate to in-game progression across rowing, cycling, and strength modes.

World of Warcraft / Final Fantasy XIV
MMO Systems Analysis 2016 - Present
- Study class design, talent structures, progression systems, and long-term reward loops across major expansion eras.
- Analyze how class fantasy, pacing, and combat feel shape player satisfaction and expression.

Dungeons & Dragons / Pathfinder
Dungeon Master and Systems Designer 2017 - 2025
- Ran weekly homebrew campaigns for 6 to 8 players with custom encounters, reward systems, and class interactions.
- Created custom rules, classes, items, and conversion systems between D&D 5e and Pathfinder 2e.

Daggerheart
Playtester Spring 2024 - Summer 2024

- Selected for Daggerheart playtesting and delivered structured analysis grounded in play experience and mechanic breakdown.

SKILLS

Research: Study coordination, orthopaedic and biomechanics research, motion capture and gait analysis, research instrumentation and hardware, inverse kinematics, data analysis
AI / LLM: Prompt engineering, system prompt design, guardrails, RAG, LLM output evaluation, multi-provider LLM APIs (OpenAI, Anthropic, Google Gemini), agentic workflows, AI-assisted development
Programming: Python, C#, C++, Lua, MATLAB, Arduino / embedded, Git and GitHub, Perforce
Engines / Tools: Unreal Engine 5 (Blueprint + C++), Unreal gameplay AI (behavior trees, blackboards), UEFN, Unity, Roblox (Luau), OptiTrack, motion capture pipelines, .c3d workflows
Design: Systems design, combat and progression design, reward loops, class identity, encounter design, playtesting

CERTIFICATIONS & MICROCREDENTIALS

Endless Games and Learning Lab microcredential - Motion Capture for Martial Arts